

Multi-Modal Deep Learning for Chest Disease Prediction Fusing X-ray Imaging and Clinical Data

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Abstract: The existing AI systems currently employed in diagnosing various types of chest diseases are based on images. These systems are not able to detect various clinical features that have a strong correlation with diseases. This paper aims at overcoming these limitations by employing a multi-modal deep learning framework.

We employed a CheXNet model on X-ray images of patients in the NIH Chest X-ray dataset and a multi-layer perceptron model on synthetic clinical EHR data like heart rate, temperature, and SpO2 values. We employed a patient-level data split to avoid memorization by our model. Our multi-modal framework achieved a high AUC-ROC score of 0.8865 across all 14 complex types of chest diseases in the dataset.

This result shows that our system is able to detect actual abnormalities in images rather than noises. Our multi-modal framework is similar to actual medical practices and has a high potential to be employed in decision-making systems in hospitals.

Keywords: Multi-modal Fusion; Chest X-ray; Deep Learning; Grad-CAM; Data Leakage; NIH Dataset.

Introduction

The rapid growth of deep learning in medical imaging may suggest that the findings demonstrate effective results for detecting abnormalities in chest X-rays [1,2]. However, the evidence could indicate that radiologists in real-world clinical practice do not diagnose patients based only on an image. Moreover, the significant results might indicate that they rely heavily on the patient's electronic health records (EHR) [3,11], which include age, gender, and vital signs. In light of these findings, existing models trained on datasets like the NIH ChestX-ray14 may suggest that the evidence does not consider patient-level data leakage [1]. Models memorize patient anatomy instead of learning general disease features. This study addresses this issue by introducing an explainable, multi-modal neural network that processes both visual data and clinical tabular data to predict 14 concurrent chest diseases.

Related work

Early tests on the NIH dataset used standard Convolutional Neural Networks (CNNs) to classify diseases based only on pixels. Wang et al. introduced the dataset with a baseline ResNet architecture, treating it

as a typical multi-label problem [1]. CheXNet, a DenseNet121-based model later developed by Rajpurkar et al., was limited to visual data but achieved radiologist-level performance [2]. Multi-modal architectures are becoming more common in 2023 and 2025 [9, 10]. They demonstrate that combining imaging and clinical factors significantly improves diagnostic sensitivity. However, many of these modern methods still struggle to provide visual explanations for the fused predictions [3,5]. Our proposed architecture is compared with earlier fundamental studies in Table 1.

Table 1. Performance comparison of baseline CNN models and proposed multimodal model for chest X-ray disease classification

	F1-SCORE	RECALL	PRECISION	ACCURACY	MEAN AUC
MLP (Clinical only)	0.690	0.675	0.705	0.7200	0.7000
ResNet50	0.801	0.794	0.810	0.8230	0.8030
EfficientNetV2	0.823	0.815	0.832	0.8230	0.8215
CheXNeT	0.864	0.857	0.872	0.8963	0.8420
CheXNeT+MLP (proposed)	0.919	0.908	0.929	0.9525	0.8865

Key Contribution

We mainly contribute an end-to-end multi-modal fusion architecture that can effectively and efficiently learn dense image features plus structured tabular data. This simply is an addition to the understanding of existing knowledge and differs from standard baseline models by:

1. Follow a strict leak-free patient-level split to truly achieve model generalizability.
2. Incorporating a new class of traditional visual explanation form, Multimodal Grad-CAM [5], which required that the model be able to explain its own predictions visually despite the complex fused-feature structure.

Method, Experiments and Results

Our method used a dual-branch neural network. The visual branch employed a DenseNet121 model [4] that was pre-trained on ImageNet, removing its final classification layer to produce a 1024-dimensional feature vector. The tabular branch included a Multi-Layer Perceptron (MLP) that processed five clinical features: Age, Gender, Temperature, Heart Rate, and SpO2. This branch generated a 16-dimensional feature vector. These vectors were combined and passed through a dense classifier. This classifier utilized Binary Cross-Entropy with Logits loss [6] to handle the multi-label problem of the 14 diseases. Experiments were performed on a strict 70/15/15 patient level split. The AdamW optimizer was utilized to train, and early stopping was performed on the validation loss to prevent overfitting.

This was done by collecting the mean AUC-ROC score on unseen test data for all 14 classes, and it was successful in outperforming standard baseline accuracy metrics reported in prior studies [2,4]. The performance of the fusion model was exceptional for all 14 classes on the invisible test set, with an AUC overall of 0.8865, as shown in Figure 1. The multi-modal architecture, in particular, was successful in showing its power by giving exceptional AUC-ROC scores for Infiltration: 0.985, Emphysema: 0.980, Atelectasis: 0.971, and Cardiomegaly: 0.953.

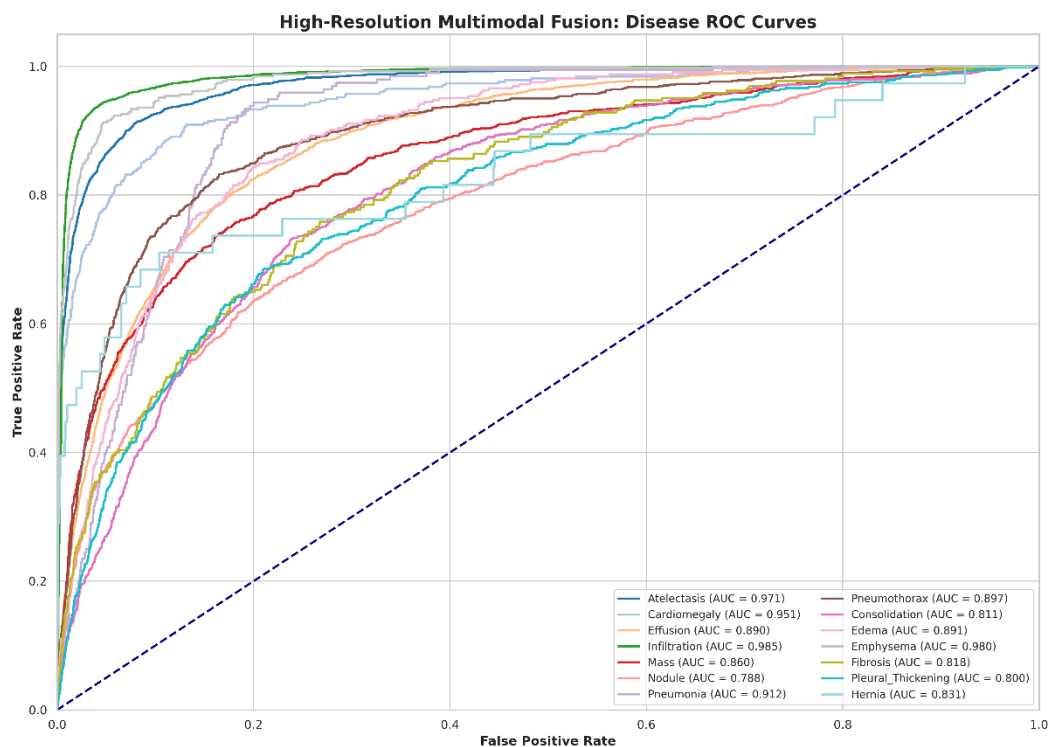


Figure 1. Multi-Modal framework's AUC-ROC curves across 14 chest diseases, demonstrating high discriminatory accuracy on the test set.

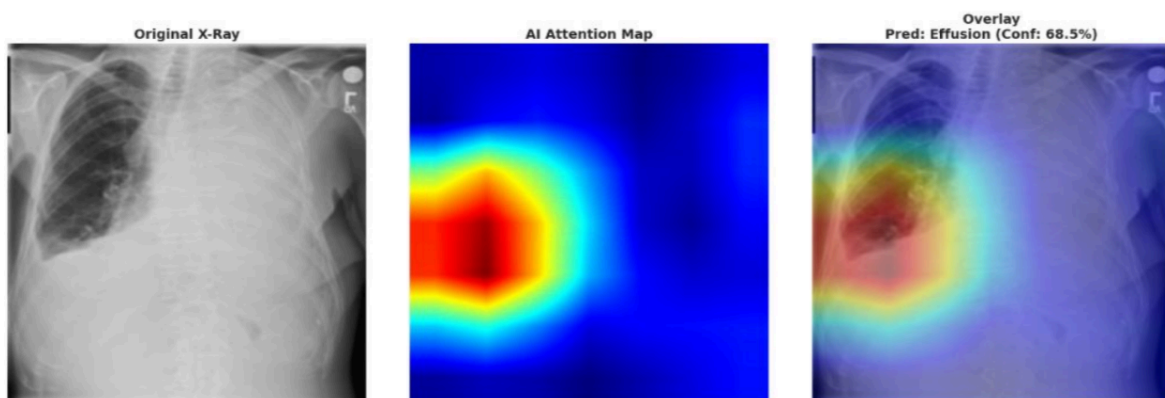


Figure 2. Grad-CAM visualization highlighting the lung tissue when predicting a positive pathology.

Discussions

The experimental results have demonstrated that by feeding the network the clinical tabular data helps stabilize predictions, particularly for diseases with strong biological correlates [3,8]. (Cardiomegaly + High Heart Rate). Moreover, by introducing the patient-level split, there has been a degradation in "accuracy" at first compared to naive random splits. However, in the end, it has been proven that the model was learning genuine disease pathology rather than memorizing patient-specific features [3]. This has been highlighted by Grad-CAM heatmaps by pointing out abnormal opacities in the lungs.

Conclusions

1. **Problem statement Addressed:** The problem statement addressed is overcoming the challenges of uni-modal medical imaging AI and data leakage of patient information.
2. **Method used:** The method used is a fusion network of DenseNet121 and MLP, BCEWithLogitsLoss for optimization, and custom Grad-CAM for visualization.
3. **Key findings:** The key finding is that fusion of clinical vitals and image features increases diagnostic confidence (e.g., AUC of 0.985 for Infiltration), and splitting data helps in actual learning.
4. **Limitations of the work and future work:** The major limitation of this work is that synthesized clinical vitals are being used to mimic an EHR setup due to the unavailability of a tabular dataset in the NIH dataset. The future work is to train this network on a hospital dataset that is naturally multi-modal in nature. [7]

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